

L-4 Data Management Practices and Beyond: Strengthening Macroeconomic Forecasting Foundations

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Introduction

- The ability to retrieve time series data in a manner that facilitates its accurate use and representation in published reports is fundamental.
- This requires:

  (a) proper documentation of all the data of relevance to the CB so that it is well defined in terms of its core attributes; and

  (b) a structured, network-based database system that facilitates accurate sharing of the data and its core attributes amongst CB staff.

- The basic problem is simple to define and understand but difficult to implement. Staff tend to work with a single frequency (e.g., quarterly), but the data they need is often not available at the target frequency. Tedious data transformations, which often need to be repeated, are required to build a dataset at a specific frequency.
- What are the key attributes of a series? How should series be saved on the network?

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Part I: Understanding Time-Series Objects the Importance and the Role of Metadata

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Part II: What is a Time-Series?

- A time-series is an array of numbers that can indexed by the date of the observation or its relative position in the array.
- For example, quarterly values of real GDP for the USA are shown in the Excel spreadsheet on the right.
- The first data point has a time-index of 1947Q1.

	A	B	C
1	Date	Real GDP	
2	1947Q1	2034.5	
3	1947Q2	2029	
4	1947Q3	2024.8	
5	1947Q4	2056.5	
6	1948Q1	2087.4	
7	1948Q2	2121.9	
8	1948Q3	2134.1	
9	1948Q4	2136.4	
10	1949Q1	2107	
11	1949Q2	2099.8	
12	1949Q3	2121.5	
13	1949Q4	2103.7	
14	1950Q1	2186.4	
15	1950Q2	2253	
16	1950Q3	2340.1	
17	1950Q4	2384.9	
18	1951Q1	2417.3	
19	1951Q2	2459.2	
20	1951Q3	2509.9	
21	1951Q4	2515.4	
22	1952Q1	2542.3	
23	1952Q2	2547.8	
24	1952Q3	2566.2	
25	1952Q4	2650.4	

Definition of Time-Series

- Note that GDP("1947Q1") = 2034.5
- Since it is in the first position of the array, GDP(1) = 2034.5
- Likewise, GDP("1952Q1") = GDP(21) = 2542.3

	A	B	C
1	Date	Real GDP	
2	1947Q1	2034.5	
3	1947Q2	2029	
4	1947Q3	2024.8	
5	1947Q4	2056.5	
6	1948Q1	2087.4	
7	1948Q2	2121.9	
8	1948Q3	2134.1	
9	1948Q4	2136.4	
10	1949Q1	2107	
11	1949Q2	2099.8	
12	1949Q3	2121.5	
13	1949Q4	2103.7	
14	1950Q1	2186.4	
15	1950Q2	2253	
16	1950Q3	2340.1	
17	1950Q4	2384.9	
18	1951Q1	2417.3	
19	1951Q2	2459.2	
20	1951Q3	2509.9	
21	1951Q4	2515.4	
22	1952Q1	2542.3	
23	1952Q2	2547.8	
24	1952Q3	2566.2	
25	1952Q4	2650.4	

Key Attributes of a Time-Series

- A time-series is an array of numbers (usually double-precision) with attributes that determine how those observations will be managed in specific situations. Key time-series attributes include:

- ***Default frequency.*** Real GDP, for example, can be *quarterly*, indicating that there will be 4 observations per year. There are several time-series frequencies, including weekly, monthly, half-yearly, 5-day week ("financial"), and daily. There is even a time-stamped frequency that allows for minutes/seconds.

- ***Default conversion method*** – this is the mathematical procedure used to move/convert the series from a higher frequency to a lower frequency, for example from quarterly to annual. Standard methods include first, last, sum, maximum, minimum or the average of the values of the period. Also, one may optionally ignore missing values when converting to a lower frequency.

Key Attributes of a Time-Series

- ***Default interpolation method*** – the mathematical procedure used to move/interpolate the series from a lower frequency to a higher frequency, for example from annual to quarterly. Standard methods include constant, linear, cubic-spline, and the Chow-Lin interpolation. Note that the same methods can be used to replace missing values in a time-series without changing its frequency.

- ***The position of the observation*** **in the period** (i.e., beginning, end, mid-point). For example, for a quarterly series, the observation may be positioned in the first, second, or third month of the quarter. The position of the observation is important when moving from a low to a higher frequency.

Time-Series Matrix

- A time-series matrix is a group of N > 1 time-series presented at a common or target frequency (e.g., quarterly).

- Each series may have a default frequency that is different from the base frequency, for example, quarterly. Typically, the default conversion/interpolation method of the series is used to move the time-series to the target frequency before constructing the time-series matrix.

Time-Series Matrix

- Each series need not have the same date range. There could be missing values at the beginning or end of the time-series matrix.

- For statistical analysis, one often works with the 'common overlap' (defined next slide) of the series such that there are no missing values at the beginning nor end of the time-series matrix.

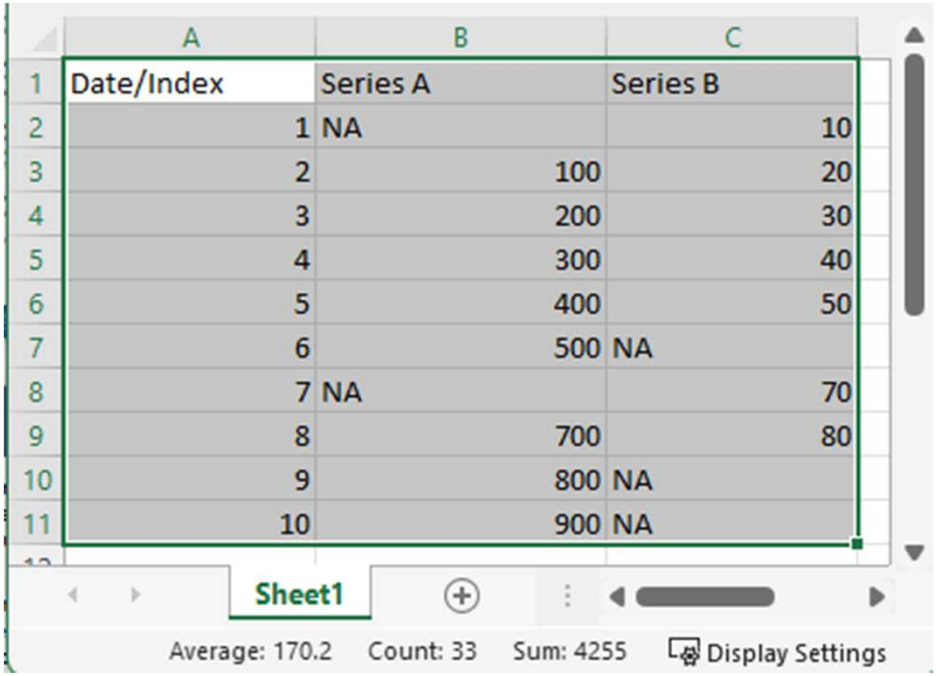
Time-Series Matrix with 2 Series

Consider the time-series matrix shown on the right. It has 2 series, each with 10 observations, some of which are missing (i.e., NA).

The common overlap involves the observations in rows 2 to 8 inclusive (missing values within the period are allowed, but not at the end or beginning of the period).

Notice that, starting from the first row (1) and working down, the first row that does not contain a missing value is row 2.

Starting from the last row (10) and working up, the last row that does not contain a missing value is row 8.



	A	B	C
1	Date/Index	Series A	Series B
2	1	NA	10
3	2	100	20
4	3	200	30
5	4	300	40
6	5	400	50
7	6	500	NA
8	7	NA	70
9	8	700	80
10	9	800	NA
11	10	900	NA

Multivariate statistical procedures would invariably base their analysis on the observations in rows 2 to 8 inclusive, unless some statistical procedure is used to replace the missing values in rows 1(Series A) and rows 9 and 10 (Series B).

Flow versus Stock Variables

- Time-series variables are often classified according to whether they are 'flow' or 'stock/index' variables.

- Flow variables are variables that **start at zero** at the beginning of each period (e.g., a year). The final value at the end of the period (e.g., a year) measures the amount of the item/good produced during a given period.

- For example, if a time-series is quarterly and a flow variable (e.g., real GDP), the value of real GDP would necessarily be zero at the beginning of the year (i.e., January 1). At the end of the year, December 31, the annual value of real GDP would be the sum of its 4 quarterly values for the year (i.e., total output for the year).

Flow versus Stock Variables

- By contrast, stock variables measure the amount of an item (e.g., the money supply) at a particular point in time. It will **not** be zero at the beginning of each period.

- Stock variables are normally positioned at the end of the period.

- Researchers typically use the sum or average of the higher frequency observations within the period to convert a flow variable from a higher to a lower frequency.

- There are far more options to consider in the case of flow variables. One can use the first observation of the period, the last, the mid-point, the average, maximum, minimum, or even the median. The position of the converted flow variable with the lower frequency time frame depends on the method used to convert the series to the lower frequency.

Part II: Why do Central Banks Need a Centralized Database?

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Why do CB need a Centralized Database

- Central banks rely on increasingly complex datasets
- Fragmented systems → inconsistent numbers, slow (inefficient) workflows
- Need for secure, governed, scalable data infrastructure
- One that supports monetary policy, financial stability, macro-prudential analysis

Existing Challenges in Many Central Banks

- Data silos across departments
- Multiple versions of the same dataset
- Manual Excel workflows
- Limited metadata and poor auditability
- High operational risk

Why Centralization Is Important

- Single source of truth
- Consistent numbers across the institution
- Automated import and validation
- Full traceability and reproducibility
- Strong security and access control
- Robust storage
- Industry standard, automated backup procedures.

Essential Features

- Unified data model (TS, XS, and Panel)
- Rich metadata catalog
- Versioning & vintages
- High-performance query engine

Essential Features

- Automated **Extract, Transform, Load** (ETL) /ELT pipelines (e.g., having the ability to produce an appropriately aligned data matrix at any target frequency from series with different frequencies)

- Interoperability with Python, R, Stata, EViews, Excel (the database system must have an API that is directly accessible from any of these packages). This is critical for producing regular reports and research for management.

Metadata Requirements

- Source, frequency, units
- Seasonal adjustment method
- Revision history (by whom, when)
- Transformation pipeline
- Contact department
- Release calendar

Architecture Options

- **A. Centralized SQL Warehouse (e.g., Microsoft SQL server, Oracle, IBM)**
- **B. Data Lake + Metadata Layer**
- **C. Hybrid (Most common)**
  - SQL for curated data
  - Lake for raw/large datasets
  - API layer for distribution

Data Model Design

- Time-series table (SeriesID, Timestamp, Value, Vintage)
- Cross-sectional table (EntityID, Date, Variables)
- Panel table (EntityID, Timestamp, Dimensions)
- Metadata tables (attributes, lineage, revisions)

Implementation Phases

- Requirements gathering (absolutely essential)
- Architecture & schema design
- ETL pipeline development
- API & access layer
- Migration of legacy datasets
- Training & adoption

Import/Export Facilities

- Python SDK (Pandas, NumPy, Statsmodels)
- R package (tidyverse, data.table)
- Stata/EViews integration
- Excel add-in for economists
- BI tools (Power BI, Tableau)
- SDMX compatibility

Governance and Risk

- Role-based access (who is allowed to access specific series and when)
- Data stewards
- Quality checks
- Audit trails
- Compliance (Basel, GDPR, confidentiality)

Strategic Benefits

- Faster policy analysis
- Reduced operational risk
- Improved transparency
- Better reproducibility
- Foundation for ML/AI analytics

Conclusion

- Centralization is a strategic transformation — not an IT upgrade.
- It strengthens policy credibility, analytical capability, and institutional efficiency.
- Every central bank needs (or will eventually need) a centralized database system.
- The earlier it is planned for and adopted, the better.

Part III: Basic Data Management Using Excel

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Basic Data Management Using Excel

- Microsoft Excel can be used for storage of time-series data
- It is possible to follow some core organizational principles to achieve a basic database structure that provides meaningful access to critical data
- The qualification ‘basic’ is critical here because Microsoft Excel worksheets are not designed for structured data and do not scale very well.

- Moreover, network latency can become a significant issue when the number of series/observations is large

- Microsoft Excel requires macros/add-ins that enable one to import/export and transform the data to the desired frequency. These are not easy to develop – need the help of an IT specialist (or AI) to write them.

Basic Time-Series Data Management Using Excel

Basic requirements for implementing a rudimentary database system using Microsoft Excel

- Maintain and enforce an enterprise-wide minimal attribute list. No series is added to the database ***unless*** all of these attributes are present.

- Ensure that heterogenous frequencies are **not** added to the **same worksheet.** Different frequencies in a single worksheet is to be avoided at all cost**.**

- Avoid linking to other worksheets in the central database at all costs. Each worksheet should be self-contained.

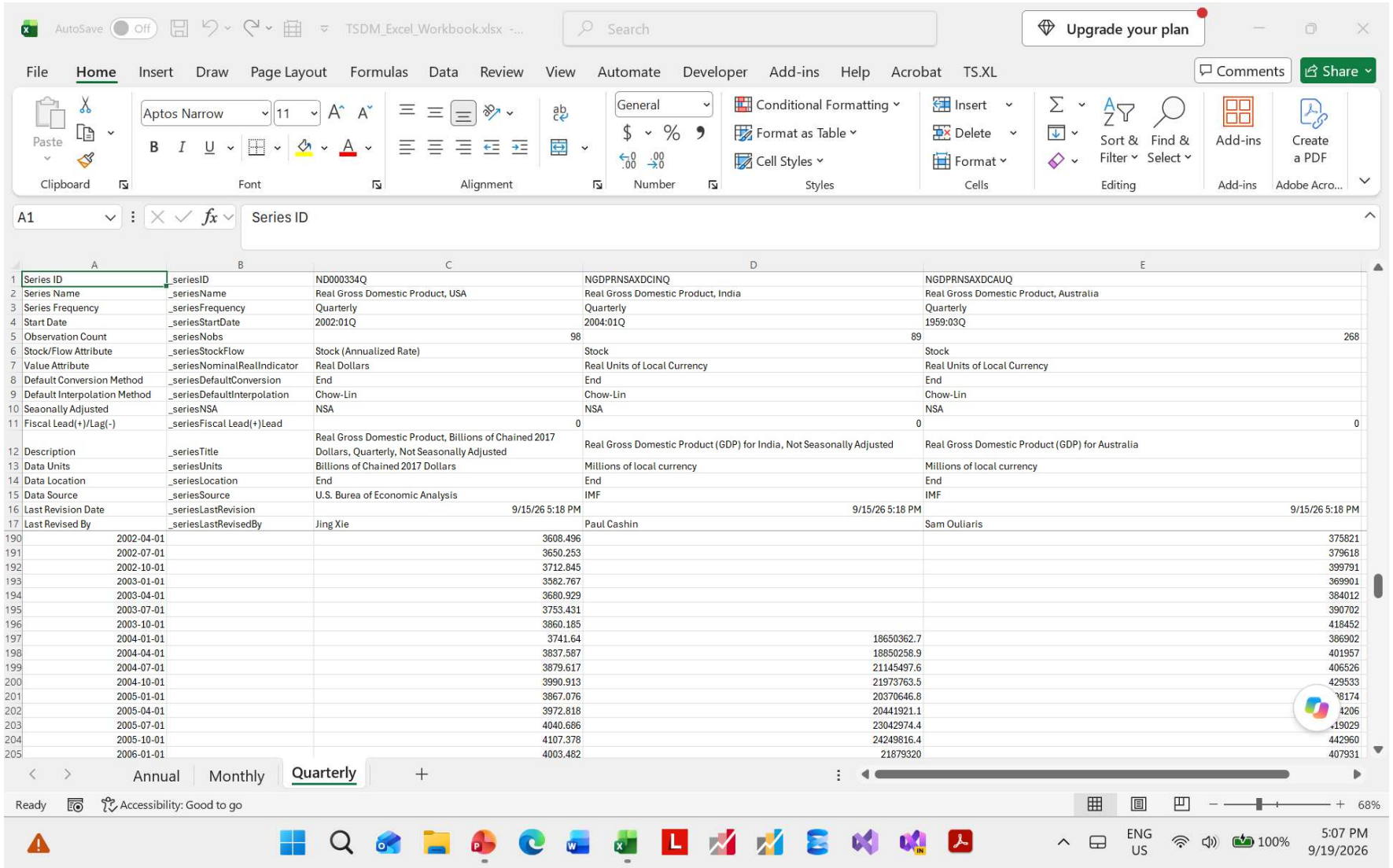
Data Management Using Excel

- Use consistent labelling for time across all worksheets. In this regard, it is desirable to adopt a **yyyy-mm-dd** dating scheme for values rather than one that depends on frequency. This makes it far easier to align the data and convert to different frequencies.

- Derived data from Excel manipulations (e.g., using an Excel function) should be in another sheet – not the same sheet as the raw data.

- Use VBA macros to build time-series matrices.

Example Excel Worksheet: TSDM_Excel.xlsx



	A	B	C	D	E
1	Series ID	_seriesID	ND000334Q	NGDPRNSAXDCINQ	NGDPRNSAXDCAUQ
2	Series Name	_seriesName	Real Gross Domestic Product, USA	Real Gross Domestic Product, India	Real Gross Domestic Product, Australia
3	Series Frequency	_seriesFrequency	Quarterly	Quarterly	Quarterly
4	Start Date	_seriesStartDate	2002:01Q	2004:01Q	1959:03Q
5	Observation Count	_seriesNobs	98	89	268
6	Stock/Flow Attribute	_seriesStockFlow	Stock (Annualized Rate)	Stock	Stock
7	Value Attribute	_seriesNominalRealIndicator	Real Dollars	Real Units of Local Currency	Real Units of Local Currency
8	Default Conversion Method	_seriesDefaultConversion	End	End	End
9	Default Interpolation Method	_seriesDefaultInterpolation	Chow-Lin	Chow-Lin	Chow-Lin
10	Seaonally Adjusted	_seriesNSA	NSA	NSA	NSA
11	Fiscal Lead(+)/Lag(-)	_seriesFiscal Lead(+)Lead	0	0	0
12	Description	_seriesTitle	Real Gross Domestic Product, Billions of Chained 2017 Dollars, Quarterly, Not Seasonally Adjusted	Real Gross Domestic Product (GDP) for India, Not Seasonally Adjusted	Real Gross Domestic Product (GDP) for Australia
13	Data Units	_seriesUnits	Billions of Chained 2017 Dollars	Millions of local currency	Millions of local currency
14	Data Location	_seriesLocation	End	End	End
15	Data Source	_seriesSource	U.S. Burea of Economic Analysis	IMF	IMF
16	Last Revision Date	_seriesLastRevision	9/15/26 5:18 PM	9/15/26 5:18 PM	9/15/26 5:18 PM
17	Last Revised By	_seriesLastRevisedBy	Jing Xie	Paul Cashin	Sam Ouliaris
190	2002-04-01		3608.496		375821
191	2002-07-01		3650.253		379618
192	2002-10-01		3712.845		399791
193	2003-01-01		3582.767		369901
194	2003-04-01		3680.929		384012
195	2003-07-01		3753.431		390702
196	2003-10-01		3860.185		418452
197	2004-01-01		3741.64	18650362.7	386902
198	2004-04-01		3837.587	18850258.9	401957
199	2004-07-01		3879.617	21145497.6	406526
200	2004-10-01		3990.913	21973763.5	429533
201	2005-01-01		3867.076	20370646.8	8174
202	2005-04-01		3972.818	20441921.1	4206
203	2005-07-01		4040.686	23042974.4	19029
204	2005-10-01		4107.378	24249816.4	442960
205	2006-01-01		4003.482	21879320	407931

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Part IV: Basic Data Management Using EViews

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Basic Data Management Using EViews

- EViews is fundamentally an econometric package written by economists for economists (meaning that the designers have an excellent understanding of how applied economists work)

- It has several functions for aligning data across time and a powerful, robust workfile/database system to support the storage/retrieval of economic data.

Basic Data Management Using EViews

- This means that it can be used to build a centralized database system for economic data, especially for small teams/departments.

- Its main disadvantage, however, is the cost of having enterprise-wide access to EViews. The per-cost of seats could be prohibitive if a truly centralized (unlimited access) system is required.

Design Principles Using EViews: Similar to Excel

- Basic requirements for implementing a database system using EViews
  - Maintain and enforce an enterprise-wide minimal attribute list.
  - No series is added to the database unless these attributes are present.
  - Avoid linking to other pagefiles at all costs.

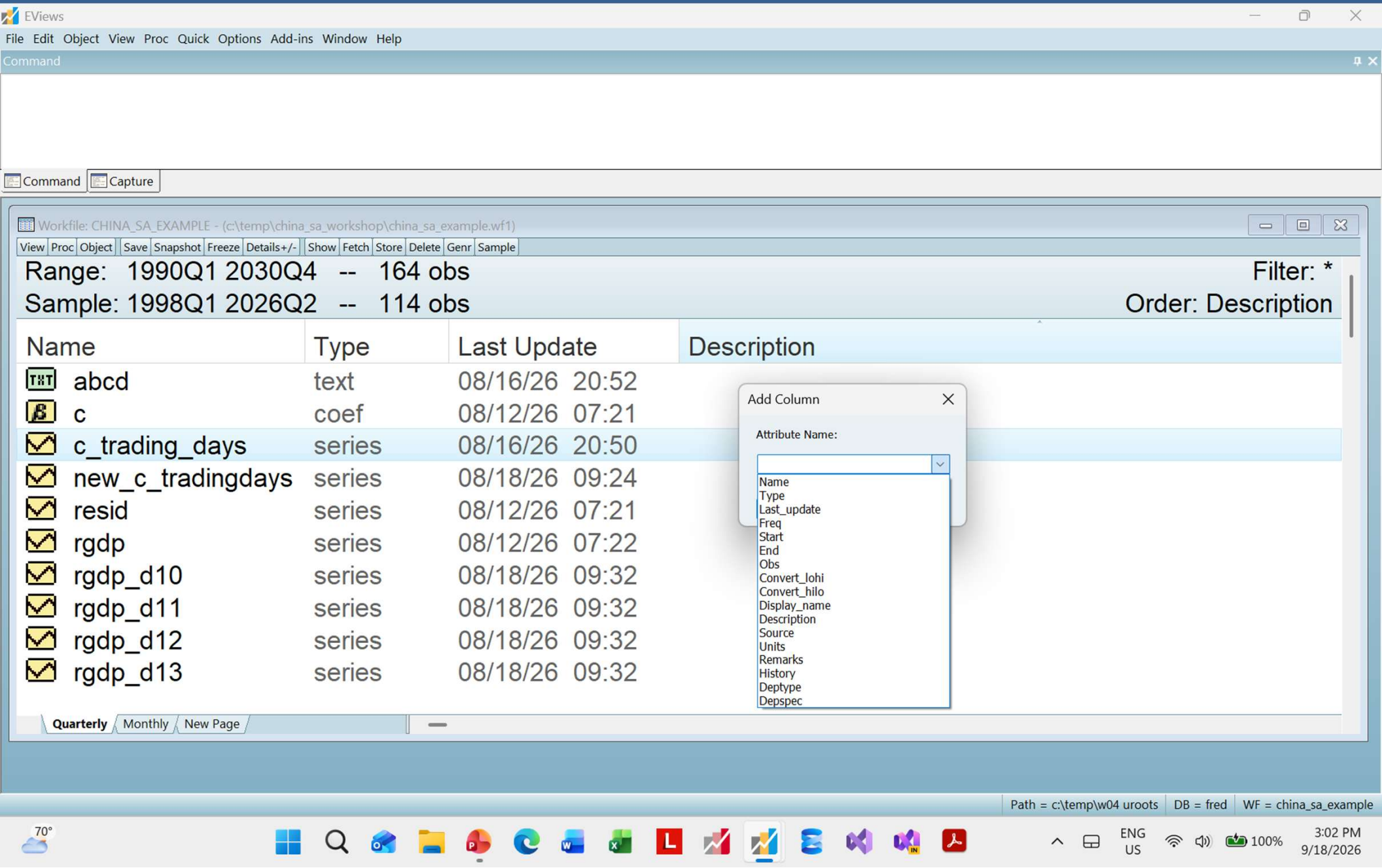
- Derived data from EViews manipulations (e.g., using an EViews function or equation) should be done using an EViews program and saved to a dedicated pagefile – never the same pagefile as the raw data.

- Develop EViews programs to build standard time-series matrices. This is relatively easy when the minimal attribute list is respected.

Adding Default Attributes to EViews Series

- EViews allows one to attach attributes to any object in the database.
- This feature can help to build a well-structured time-series database system.
- There many, so-called built-in attributes, including
  - Default conversion method (“conv_hilo”)
  - Default interpolation method (“conv_lohi”)

Built-In Attributes Supported By EViews

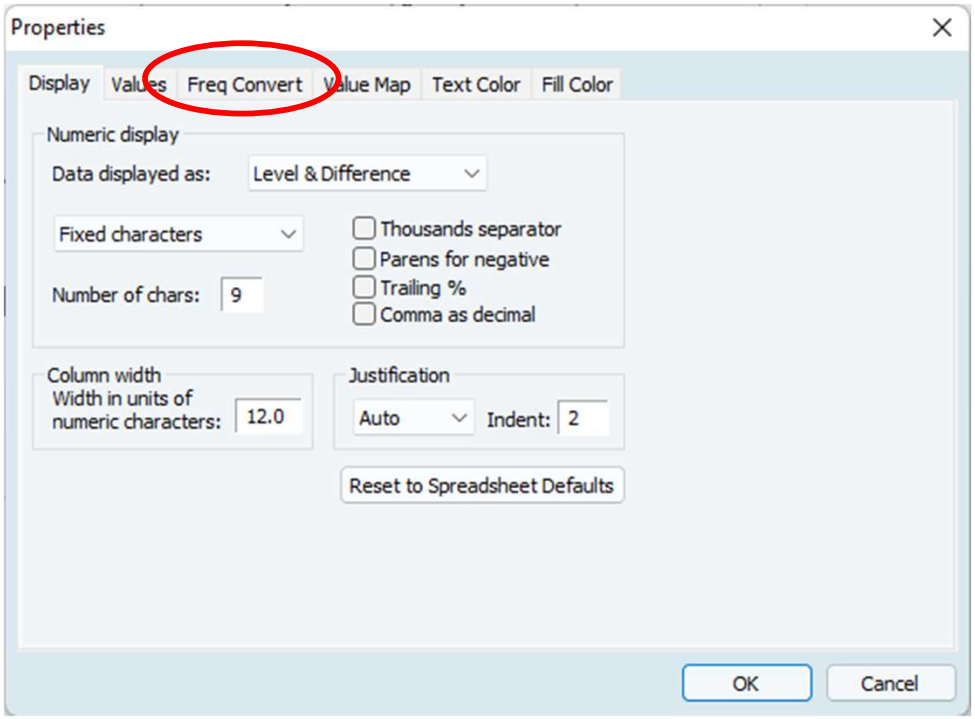


Specifying Default Conversion Methods

It would be tedious to carry out copy/paste operations and have specify the various options each time a frequency conversion is required.

When performing frequency conversion of one or more series, EViews can refer to the attributes of the series to perform the conversion. These settings may be specified using the **Freq Convert** tab of the **Properties** dialog of the series (accessible from Spreadsheet View).

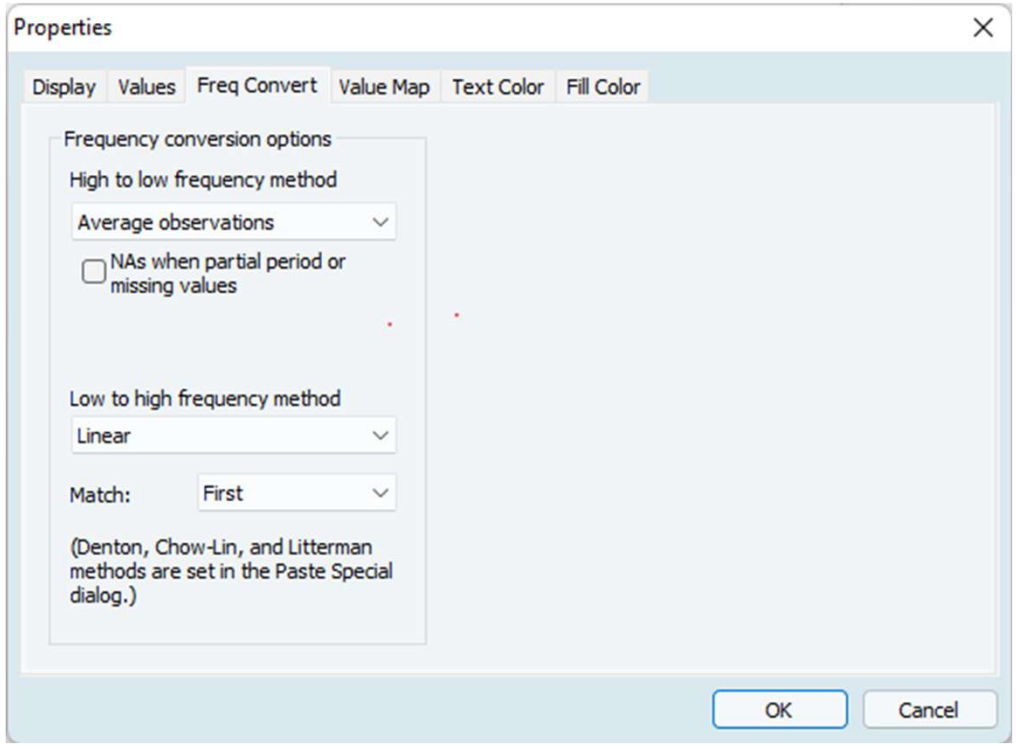
Click on the **Properties** button on the series toolbar and select the **Freq Convert** tab to access the dialog box.



Specifying Default Conversion Methods

The series properties dialog box allow you specify the default conversion methods to be used when moving from high to low (e.g., Average) frequency or from low to high (e.g., Linear)

This enables you to do frequency conversion by copying and pasting series from one pagefile to another with a different frequency.



These Attributes Can Be Easily Managed Using EViews Code

- The **.setconvert** attribute allows you to set the default conversion and interpolation methods for a series.

- For example, the command:

  rgdp.**setconvert** l c

  sets the default conversion method for converting the series **rgdp** to a lower frequency to "l" (or "Last", ignoring missing values), and "c" (cubic spline) for converting from low to a higher frequency (i.e., interpolation).

- The command:

  rgdp.**setconvert** ln c

  allows missing values to propagate during consolidation of **rgdp** to a lower frequency.

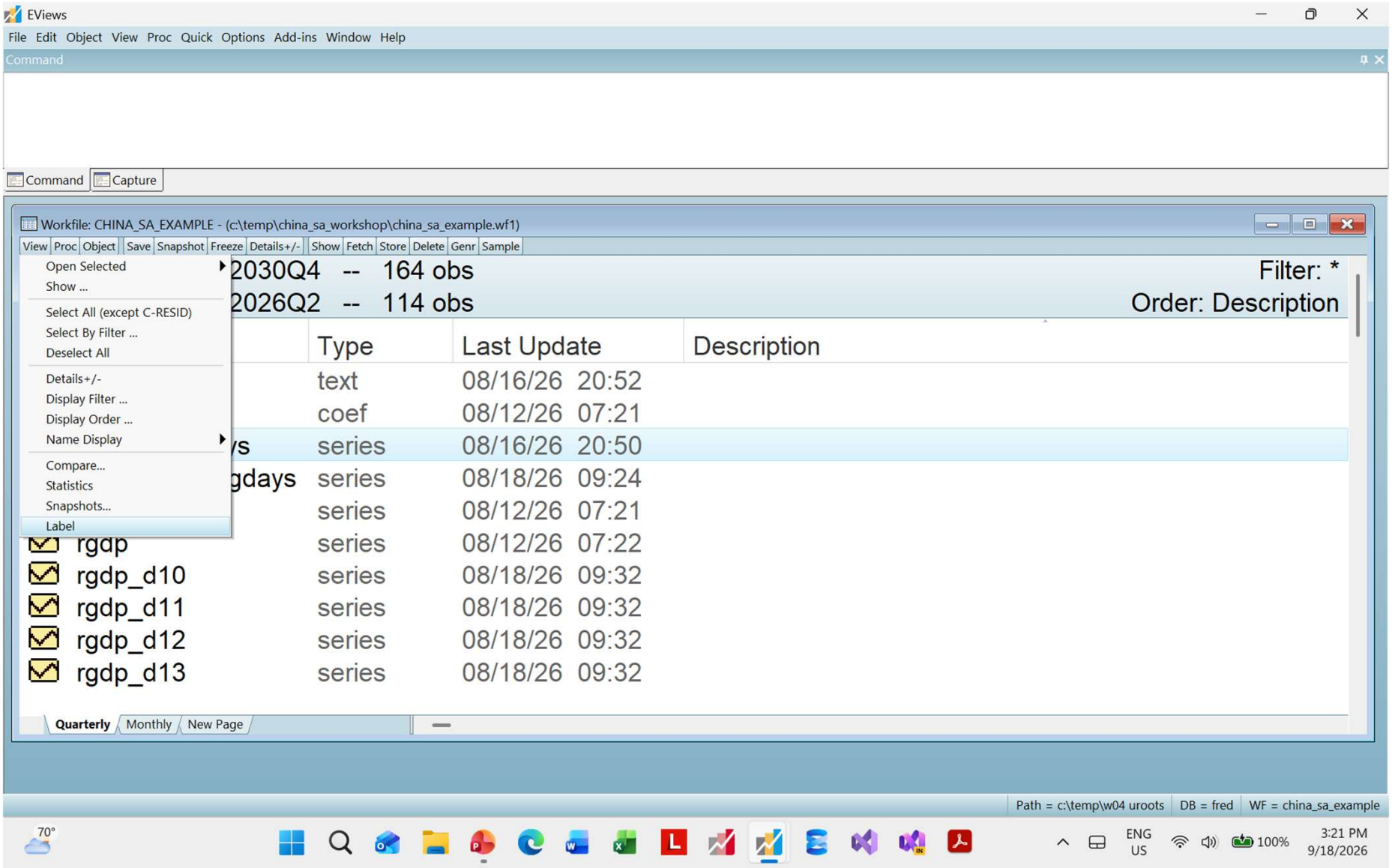
Adding Custom Attributes

- EViews also allows you to add custom attributes. This is where you can adapt their workfile to your own needs.

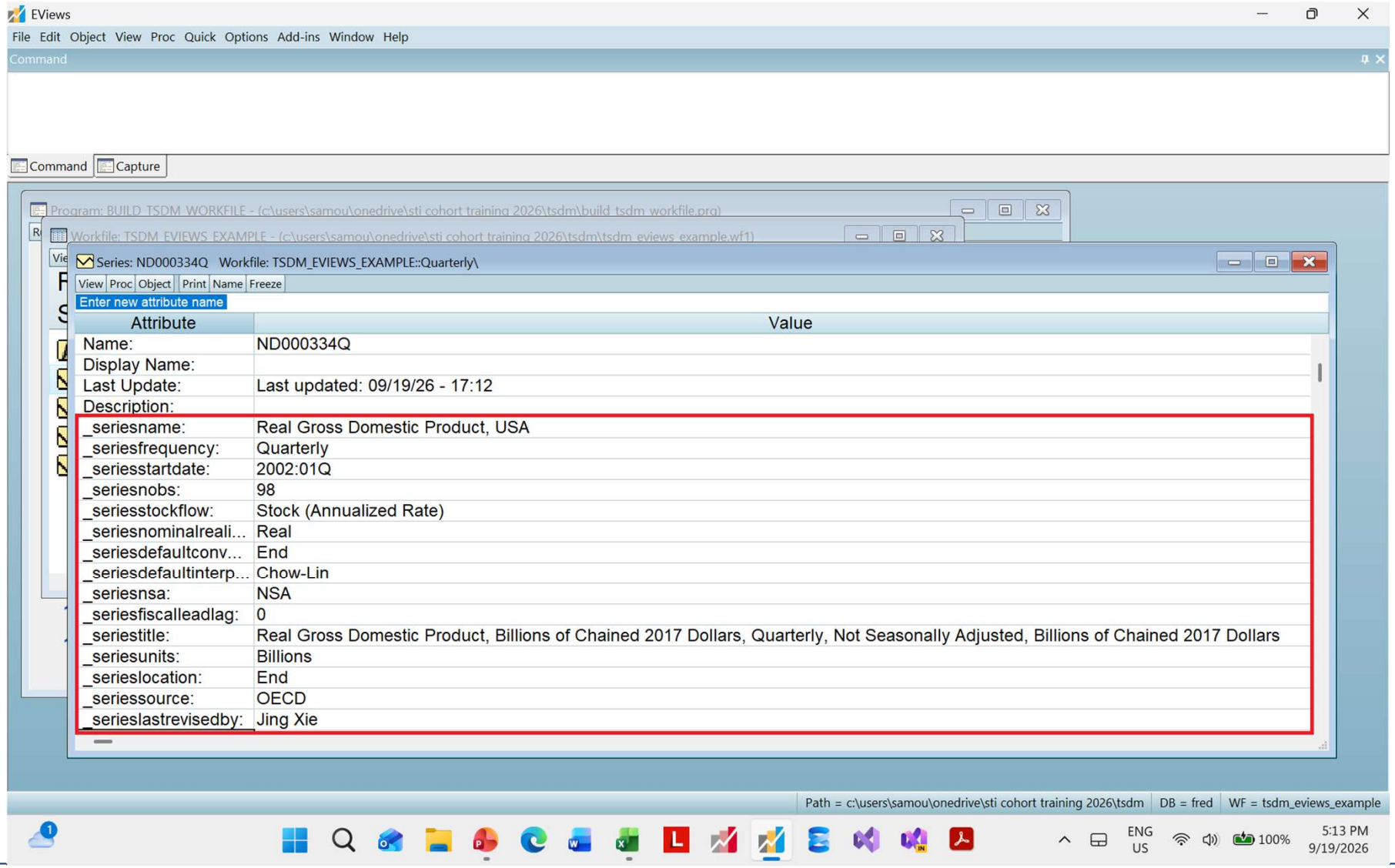
- Custom attributes can be added manually using the View -> Label command

- Simply click on any series and invoke View -> Label.

Manual Addition of Custom Attributes



Manual Addition Of Custom Attributes



Addition of Custom Attributes: Using an EViews Program

- EViews allows you to set custom attributes programmatically. The EViews command is ".setattr" and its syntax is:

  series_name.setattr(custom_attr) attr_value

- This sets the attribute "custom_attr" to "attr_value".

- Once added to a series, the attribute may be extracted using the @attr data member.

- For example, the command sequence:

```
a.setattr(_seriesrevised) 2026/09/01 14.22.06
string s = a.@attr("_seriesrevised")
```

sets the "**_seriesrevised**" attribute in the series object "**a**" to the string "2026/09/01 14.22.06", and extracts the attribute into the string object **s**.